$2x+y=3$ — (I)

$x-2y=-1$ — (II)

$x=2y-1$

4 unknown 3eqn

row picture.

$2(2y-1)+y=3$

$5y=5$

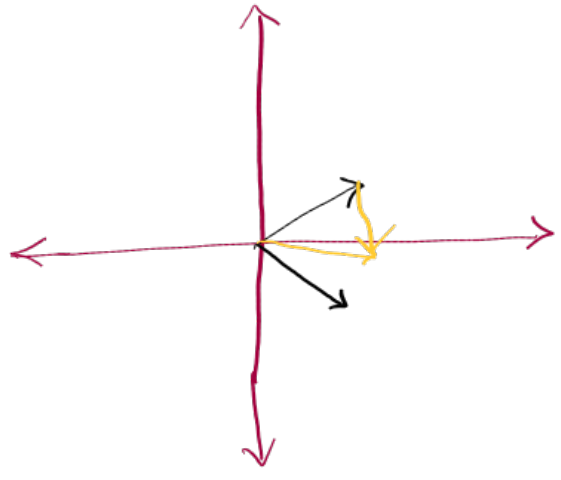
$y=1$

$2x+y=3$

$x-2y=-1$

$\begin{bmatrix}2\\1\end{bmatrix}x+\begin{bmatrix}1\\-2\end{bmatrix}y=\begin{bmatrix}3\\-1\end{bmatrix}$

column pictures: differentiating them to columns



Rewrite as a matrix:

$$\begin{bmatrix}2 & 1\\1 & -2\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}3\\-1\end{bmatrix}$$

A x b

linear transformation on an unknown vector. Input unknown Output known.

$A^{-1}Ax=A^{-1}b$

$x=A^{-1}b$

$$2x - y = 0$$
$$-x + 2y - z = -1$$
$$-3y + 4z = 4$$

$$\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} x + \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} y + \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} z = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}}_{b}$$

* Can we solve $Ax = b$ for all b?

=> possible if columns are linearly independent of one another.